

Project Initialization and Planning Phase - Proposed Solution

Date	29 September 2026
Team ID	6ab3e6756603573ccfb3b891
Project Name	FitBuddy - AI Fitness Plan Generator Using Gemini Models
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

Project Overview

Objective	Provide a simple AI-powered way to generate a personalized 7-day workout plan and nutrition/recovery tip from basic user details.
Scope	User profile form, AI workout generation, nutrition tip generation, feedback-based plan update, SQLite storage, result display and admin users view.
Problem Statement	Generic plans can be difficult to personalize and modify. Users need a structured weekly routine that reflects their goal and intensity.
Impact	Reduces manual planning effort, gives an editable starting point and demonstrates practical Generative AI integration in a web application.
Proposed Approach	Collect profile -> validate -> call Gemini -> store -> display -> accept feedback -> regenerate updated plan.
Key Features	Personalized 7-day plan; nutrition/recovery tip; feedback loop; database storage; admin view; API docs; responsive frontend.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computer / laptop	4 GB RAM minimum; internet connection for Gemini API
Software	OS + development tools	Windows/Linux/macOS, Python, VS Code, web browser
Libraries	Application stack	FastAPI, Jinja2, SQLAlchemy, Pydantic, google-genai, Pytest
Data	User input and generated plans	SQLite database with users and plans tables
API credential	Gemini access	GEMINI_API_KEY stored in .env